POWER BI

INTERVIEW QUESTIONS & ANSWERS

A practical revision guide for beginners and aspiring data analysts

FUNDAMENTALS Core concepts and components	**DATA MODELING** Relationships and star schema
DAX Measures and calculation context	**REAL SCENARIOS** Troubleshooting and dashboard design

Includes concise answers, scenario questions, and useful DAX examples.

Interview tip: explain the concept first, then give a short example from a report or dataset.

1. Power BI Fundamentals

Q. What is Power BI?

Answer: Power BI is Microsoft's business intelligence tool used to connect to data, transform it, build data models, and create interactive reports and dashboards.

Q. What are the main components of Power BI?

Answer: Power BI Desktop, Power BI Service, Power BI Mobile, On-premises Data Gateway, and Power BI Report Builder are key components.

Q. What is Power BI Desktop?

Answer: A Windows application used to connect to data, transform it with Power Query, create relationships and measures, and design reports.

Q. What is Power BI Service?

Answer: The cloud service where users publish, share, collaborate on, and manage Power BI content.

Q. What is a report?

Answer: A report is a collection of interactive pages containing visuals based on a semantic model.

Q. What is a dashboard?

Answer: A dashboard is a single-page canvas in Power BI Service that can contain tiles pinned from one or more reports or semantic models.

Q. What is a semantic model (dataset)?

Answer: It is the modeled collection of tables, relationships, measures, and metadata that reports use. 'Semantic model' is the newer term for what was commonly called a dataset.

Q. What data sources can Power BI connect to?

Answer: Examples include Excel, CSV, SQL Server, Oracle, SharePoint, web sources, Azure services, and many cloud applications.

2. Power Query and Data Preparation

Q. What is Power Query?

Answer: Power Query is the data connection and transformation engine used to extract, clean, reshape, and combine data before loading it.

Q. What is ETL?

Answer: ETL means Extract, Transform, Load: obtain data, prepare it, and load it into a target model or destination.

Q. What is the difference between Merge and Append?

Answer: Merge joins tables side-by-side using matching columns; Append stacks rows from tables with compatible columns.

Q. What is data profiling?

Answer: It examines column quality, distribution, and statistics to identify missing, invalid, or unusual values.

Q. What is query folding?

Answer: Query folding pushes supported transformation steps back to the source system so the source performs the work, often improving refresh performance.

Q. What is the difference between Remove Rows and Remove Duplicates?

Answer: Remove Rows deletes rows based on a rule or position; Remove Duplicates removes rows that repeat according to selected columns.

Q. What is the M language?

Answer: M is the case-sensitive formula language used by Power Query for data extraction and transformation.

Q. What is data type detection important for?

Answer: Correct data types help calculations, sorting, filtering, relationships, and time intelligence work as expected.

3. Data Modeling and Relationships

Q. What is data modeling?

Answer: Data modeling organizes tables, relationships, and calculations so data can be analyzed accurately and efficiently.

Q. What is a star schema?

Answer: A model with a central fact table connected to dimension tables; it is generally easy to understand and performs well.

Q. What is a fact table?

Answer: A table containing measurable events such as sales, quantities, costs, or transactions.

Q. What is a dimension table?

Answer: A table containing descriptive attributes such as customer, product, date, or region.

Q. What is cardinality?

Answer: It describes how rows relate between tables, commonly one-to-many, many-to-one, one-to-one, or many-to-many.

Q. What is a relationship's cross-filter direction?

Answer: It determines how filters travel between related tables, typically single direction or both directions.

Q. Why should bidirectional filtering be used carefully?

Answer: It can create ambiguous filter paths and unexpected results; use it only when the model requires it.

Q. What is a date table and why use one?

Answer: A dedicated calendar table supports consistent year, quarter, month, and time-intelligence calculations. Mark it as a date table when required.

4. DAX and Calculations

Q. What is DAX?

Answer: Data Analysis Expressions (DAX) is the formula language used for measures, calculated columns, and calculated tables in Power BI.

Q. What is the difference between a measure and a calculated column?

Answer: A measure is calculated at query time based on filter context; a calculated column is computed for each row during model processing and stored in the model.

Q. What is a calculated table?

Answer: A table created by a DAX expression and stored in the model when it is processed.

Q. What is row context?

Answer: Row context means an expression is evaluated for the current row, as in a calculated column or iterator function.

Q. What is filter context?

Answer: Filter context is the set of filters from visuals, slicers, report/page filters, and DAX expressions that affects a calculation.

Q. What does CALCULATE do?

Answer: CALCULATE evaluates an expression in a modified filter context and is one of the most important DAX functions.

Q. What is the difference between SUM and SUMX?

Answer: SUM adds values from one column; SUMX iterates over a table and evaluates an expression for each row before aggregating.

Q. What is the difference between COUNT, COUNTA, and DISTINCTCOUNT?

Answer: COUNT counts numeric/date values, COUNTA counts non-blank values, and DISTINCTCOUNT counts unique values.

Q. What does RELATED do?

Answer: RELATED retrieves a value from a related table, commonly from the one-side of a many-to-one relationship when row context exists.

Q. What is time intelligence in DAX?

Answer: Functions and patterns for calculations over dates, such as year-to-date, previous month, or same period last year.

5. Visuals, Filters, and Reports

Q. What is a slicer?

Answer: A visual control that lets report users filter data interactively.

Q. What are visual-, page-, and report-level filters?

Answer: Visual filters affect one visual; page filters affect visuals on one page; report filters affect the report's pages.

Q. What is drill-down?

Answer: Moving from a higher level of a hierarchy to a more detailed level within a visual.

Q. What is drill-through?

Answer: Opening a detail page filtered to the selected data point or category.

Q. What are bookmarks?

Answer: Saved report states that can capture settings such as page, filters, and visibility, often used for navigation or storytelling.

Q. What is a tooltip page?

Answer: A report page designed to appear as a custom tooltip when users hover over a visual data point.

Q. What is conditional formatting?

Answer: Formatting visual elements based on values or rules, such as highlighting low performance in red.

Q. How do you choose an appropriate visual?

Answer: Match the visual to the question: line charts for trends, bars for category comparisons, cards for KPIs, and tables for detailed records.

6. Power BI Service, Refresh, and Security

Q. What is a workspace?

Answer: A collaborative area in Power BI Service for organizing reports, semantic models, dashboards, and related content.

Q. What is a gateway?

Answer: A bridge that allows the Power BI Service to securely access supported data sources in a private or on-premises network.

Q. What is scheduled refresh?

Answer: A configuration that refreshes imported model data on a recurring schedule, subject to licensing and capacity limits.

Q. What is Import mode?

Answer: Data is loaded into the Power BI model, usually giving fast report interactions but requiring refresh to update imported data.

Q. What is DirectQuery?

Answer: Queries are sent to the underlying source when users interact with the report, subject to source and connector capabilities.

Q. What is a composite model?

Answer: A model that combines tables using different storage modes or sources, depending on supported configurations.

Q. What is Row-Level Security (RLS)?

Answer: RLS restricts which rows a user can see, based on defined roles and user identity.

Q. What is the difference between sharing and publishing?

Answer: Publishing sends content from Desktop to the Service; sharing grants other users access to supported content in the Service.

Q. What is incremental refresh?

Answer: It refreshes only recent or changed partitions of a large table rather than reprocessing all historical data, when configured and supported.

Q. What is the difference between a personal and standard gateway?

Answer: A personal-mode gateway is intended for an individual user's use cases; standard mode supports broader shared and enterprise scenarios.

7. Performance and Troubleshooting

Q. How can you improve Power BI report performance?

Answer: Use a star schema, remove unused columns, reduce high-cardinality fields where possible, prefer measures for aggregations, optimize DAX, and inspect Performance Analyzer results.

Q. Why can a Power BI file become large?

Answer: Large row counts, many high-cardinality columns, duplicated data, and unnecessary calculated columns can increase model size.

Q. What is Performance Analyzer?

Answer: A Power BI Desktop tool that records how long visuals take and helps identify slow report elements.

Q. What is the difference between a data refresh error and a visual error?

Answer: A refresh error prevents or interrupts updating model data; a visual error can arise from query, model, DAX, permissions, or rendering issues.

Q. Why might a total in a measure differ from adding the visible rows?

Answer: Measures are recalculated in the total's filter context; the total is not always the sum of displayed row-level results.

Q. How would you troubleshoot incorrect numbers?

Answer: Check source data, data types, relationships, filter context, DAX logic, date table setup, and whether filters are interacting as expected.

8. Scenario-Based Interview Questions

Q. Sales are duplicated after joining two tables. What would you check?

Answer: Check the join keys and relationship cardinality, confirm whether either side contains duplicate keys, and validate the intended grain before merging.

Q. A report is slow after adding several slicers. What would you do?

Answer: Use Performance Analyzer, inspect expensive visuals and DAX, simplify interactions, optimize the model, and reduce unnecessary high-cardinality fields.

Q. A manager wants sales year-to-date and last-year sales. What would you need?

Answer: A reliable date table related to the fact table, a sales measure, and suitable DAX time-intelligence measures.

Q. Users should see only their own region's data. How can you implement this?

Answer: Configure Row-Level Security with appropriate roles, potentially using a user-to-region mapping table and dynamic DAX, then test with representative users.

Q. A source has monthly files with the same columns. How would you combine them?

Answer: Use Power Query's folder connector or an equivalent combine-files workflow, standardize schemas, and validate the appended result.

Q. When would you choose Import over DirectQuery?

Answer: Choose Import when refresh latency and model size are acceptable and fast interactions are priorities; consider DirectQuery when source-level freshness or data-volume constraints make it appropriate.

Q. How would you design a sales dashboard for executives?

Answer: Clarify business goals, define KPIs, model sales/customer/product/date data, create trend and comparison visuals, add useful slicers, validate figures, and publish with appropriate security.

Q. A measure works in a card but not as expected in a table. Why?

Answer: The visual creates a different filter context or row grouping; inspect relationships, measure logic, and the fields used in the table.

9. Quick DAX Practice

Q. Total Sales

Answer: Total Sales = SUM(Sales[SalesAmount])

Q. Total Quantity

Answer: Total Quantity = SUM(Sales[Quantity])

Q. Average Order Value

Answer: Average Order Value = DIVIDE([Total Sales], DISTINCTCOUNT(Sales[OrderID]))

Q. Profit

Answer: Profit = SUM(Sales[SalesAmount]) - SUM(Sales[Cost])

Q. Profit Margin

Answer: Profit Margin = DIVIDE([Profit], [Total Sales], 0)

Q. Sales YTD

Answer: Sales YTD = TOTALYTD([Total Sales], 'Date'[Date])

Q. Previous Year Sales

Answer: Previous Year Sales = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date]))

Q. Distinct Customers

Answer: Distinct Customers = DISTINCTCOUNT(Sales[CustomerID])

Q. Dynamic Region Filter Concept

Answer: Use RLS roles and a user-to-region mapping table; the exact DAX depends on the model design.